

Motivation Letter: PhD Student High-Fidelity Hypersonic Flows (R1)

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28 July 2026

Recruitment Panel

Barcelona Supercomputing Center - Centro Nacional de Supercomputación

**Subject: Application for PhD Student High-Fidelity Hypersonic Flows (R1), reference
179_26_CASE_LSCFD_R1**

Dear Members of the Recruitment Panel,

I am writing to express my interest in the PhD position on high-fidelity hypersonic flows within BSC's Large-Scale Computational Fluid Dynamics team. My background is in mechanical and process engineering rather than aeronautical engineering, but the project's combination of turbulence modelling, machine-learning methods, numerical simulation, and high-performance computing closely matches the interdisciplinary research direction I want to pursue.

I completed a BSc in Mechanical Engineering at the Budapest University of Technology and Economics under a Stipendium Hungaricum scholarship, with a Process Engineering specialisation and a final grade of 4.33. I am now completing an MSc in Modelling for Science and Engineering at the Universitat Autònoma de Barcelona, specialising in Data Science. My BSc thesis built and validated a neural-network model for turbulent-intensity prediction from wind-profile data. That work gave me direct experience joining fluid-mechanical reasoning with data preparation, model development, validation, and careful interpretation of approximation limits.

My engineering training included ANSYS Fluent and other numerical and CAD/FEA tools. At ABS Consulting, I supported seismic qualification and structural assessment projects for UK nuclear power assets using Abaqus, Mathcad, Inventor, and STAAD.Pro. The work required controlled calculations, standards-based reasoning, reviewable technical reports, and communication with engineers working under demanding assurance requirements. These habits are highly relevant to research in which numerical accuracy, reproducibility, and physical interpretation cannot be separated.

My current work has moved deeper into AI and software systems. At Aily Labs, I have worked from prompt-defined MCP capabilities and gated skills through semantic SQL-agent access and production evaluation. The evaluation layer uses curated question sets, declared data and tool expectations, trace inspection, and LLM-as-a-judge scoring. This has strengthened my Python engineering, CI/CD, experimental design, and ability to distinguish model limitations from data or system-integration faults.

Through a JAE Intro ICU 2025 scholarship at IIIA-CSIC, my MSc thesis focuses on a modular ROS 2 architecture for interactive robotics, including symbolic grounding, structured language-model planning, deterministic execution, feedback, and controlled evaluation. Although this is a different application domain, it has reinforced my approach to machine-learning components whose outputs must remain constrained by physical or execution evidence.

My personal iTrader project provides a more training-intensive research environment. Its implemented RLVR-style baseline uses market-window simulation, validated task specifications, PPO/GAE, task-conditioned FiLM networks, and finite decomposed rewards. The longer-term architecture introduces capability-profiled solver heads, grouped proposer optimisation, and latent Market-JEPA representations before progressing through replay, shadow mode, deterministic risk, and eventual live execution. The domain differs from CFD, but the methodological concerns are shared: scale-sensitive objectives, distribution shift, reproducible runs, and evaluation beyond a single aggregate score.

I want to be precise about my present fit. I have used CFD software and studied turbulence-related machine learning, but I have not yet developed an in-house hypersonic CFD solver or led exascale GPU simulations. My direct GPU-porting and large-scale HPC experience is limited. I therefore view this position as a PhD training path into high-Mach LES, wall modelling, GPU-aware numerical methods, and scientific software, not as an area in which I already claim specialist status. I would be grateful for the panel's consideration of whether my mechanical/process engineering degree is accepted as equivalent to the listed aeronautical background.

What I would bring is a strong mathematical and physical foundation, practical engineering discipline, experience applying machine learning to a turbulence problem, and sustained evidence of learning across fields. I have studied and worked in multinational environments in Hungary, the United Kingdom, and Spain, and I work fluently in English and Spanish. I am motivated by collaborative research and by the opportunity to contribute to publishable work that connects physical models, numerical methods, machine learning, and modern computing architectures.

BSc's LS/CFD programme offers exactly the environment in which these interests can be developed rigorously. I would welcome the opportunity to discuss my eligibility and how my background could contribute to the high-speed research activities of the group.

Yours sincerely,

Juan David Bendek Williamson

References

1. **Roger Montane**, Mid Data Scientist, Aily Labs

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